

Image Super-Resolution Via a Convolutional Neural Network

CS229 Machine Learning • Deep Learning Reproduction & Residual Enhancement

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Motivation and Overview

Image super-resolution attempts to counteract the effects of faulty imaging hardware or software degradation by interpolating between pixels. A super-resolution convolutional neural network (SRCNN) uses a sequence of convolutional layers—a feature extraction layer, a non-linear mapping layer, and a feature reconstruction layer—to relate patches of low-resolution pixels to higher-resolution improvements.

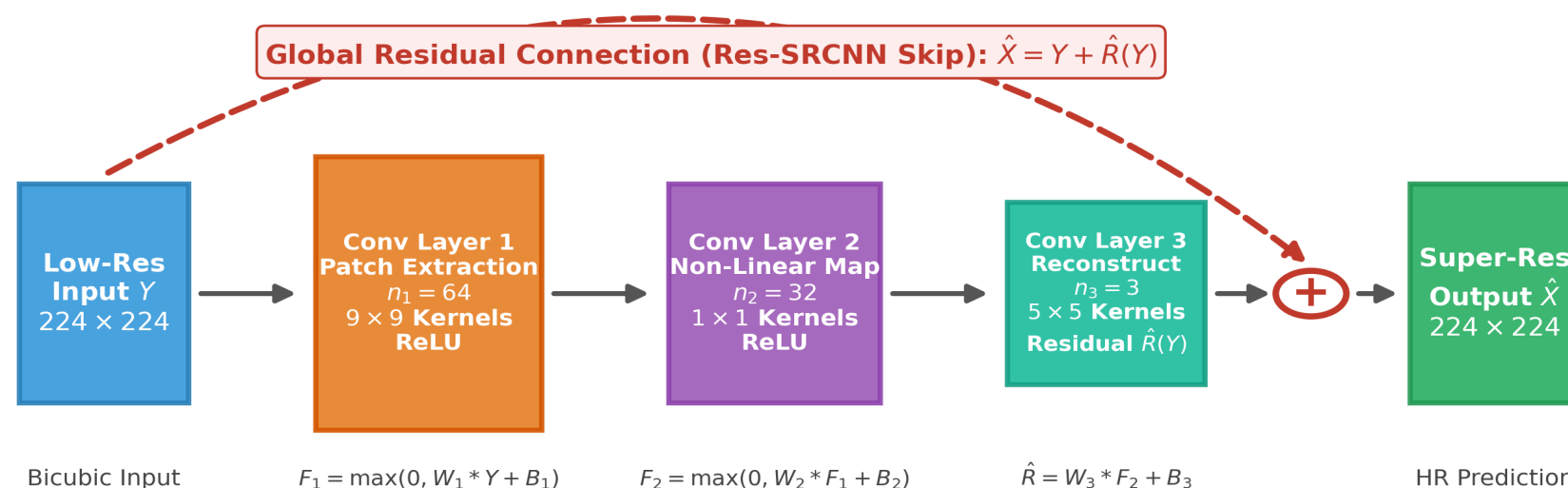


Figure 1. SRCNN architecture with Global Residual Skip Connection

The SRCNN which we are using was proposed by Dong et al. [1] and is a 3 hidden layer deep neural network. Patch extraction uses $n_1 = 64 \times 9 \times 9$ kernels, non-linear mapping uses $n_2 = 32 \times 1 \times 1$ kernels, while reconstruction uses $n_3 = 3 \times 5 \times 5$ kernels. In our implementation, a global residual identity connection directly bridges bicubic input to output, empowering the network to learn only high-frequency detail residuals.

Dataset: DIV2K

- **Super-resolution specific dataset** of large, high-definition photographs (typical native size 2040×1400 px).
- **100 curated photographs**: 60 training (60%), 20 validation (20%), and 20 test (20%).
- Contains rich high-frequency structures spanning architecture, nature, textures, and fine geometric patterns.

Preprocessing

- **Spatial Standardization**: Center-crop native images to 800×800 px for uniformity, then resize to 224×224 px.
- **Degradation Pipeline**: For network inputs, downsize by 0.5 (2x factor) using bilinear interpolation, then upsample back to 224×224 px.



Figure 2. Example of downsampled input image (left) and target image (right)

Experiments

- **Evaluation Metric**: Peak Signal-to-Noise Ratio: $PSNR = 10 \cdot \log_{10}(MAX^2 / MSE)$ and SSIM.
- **Baselines**: Nearest neighbor, bilinear, bicubic upsampling:
 - Bicubic achieves highest baseline test PSNR: **28.63 dB** (MSE: 0.00179, SSIM: 0.8815)
 - Bilinear baseline test PSNR: **27.46 dB**, Nearest Neighbor: **26.67 dB**
- **Learning Rate Optimization**: Learning rate of **0.001** was most effective:
 - Highest test PSNR: **29.46 dB** with residual learning (+0.83 dB over bicubic)
 - Learning rate LR = 0.01 showed loss instability; LR = 0.0001 converged too slowly
- **Residual Learning Innovation (Res-SRCNN)**:
 - Global skip connection allows network to predict high-frequency residual detail rather than raw image
 - Accelerates convergence from >50 epochs to 15 epochs, drastically stabilizing gradient flow
- **Training Protocol**: Adam optimizer ($\beta_1=0.9$, $\beta_2=0.999$), batch size 8, Mean Squared Error loss.

Number of Epochs vs. PSNR for Hyperparameters

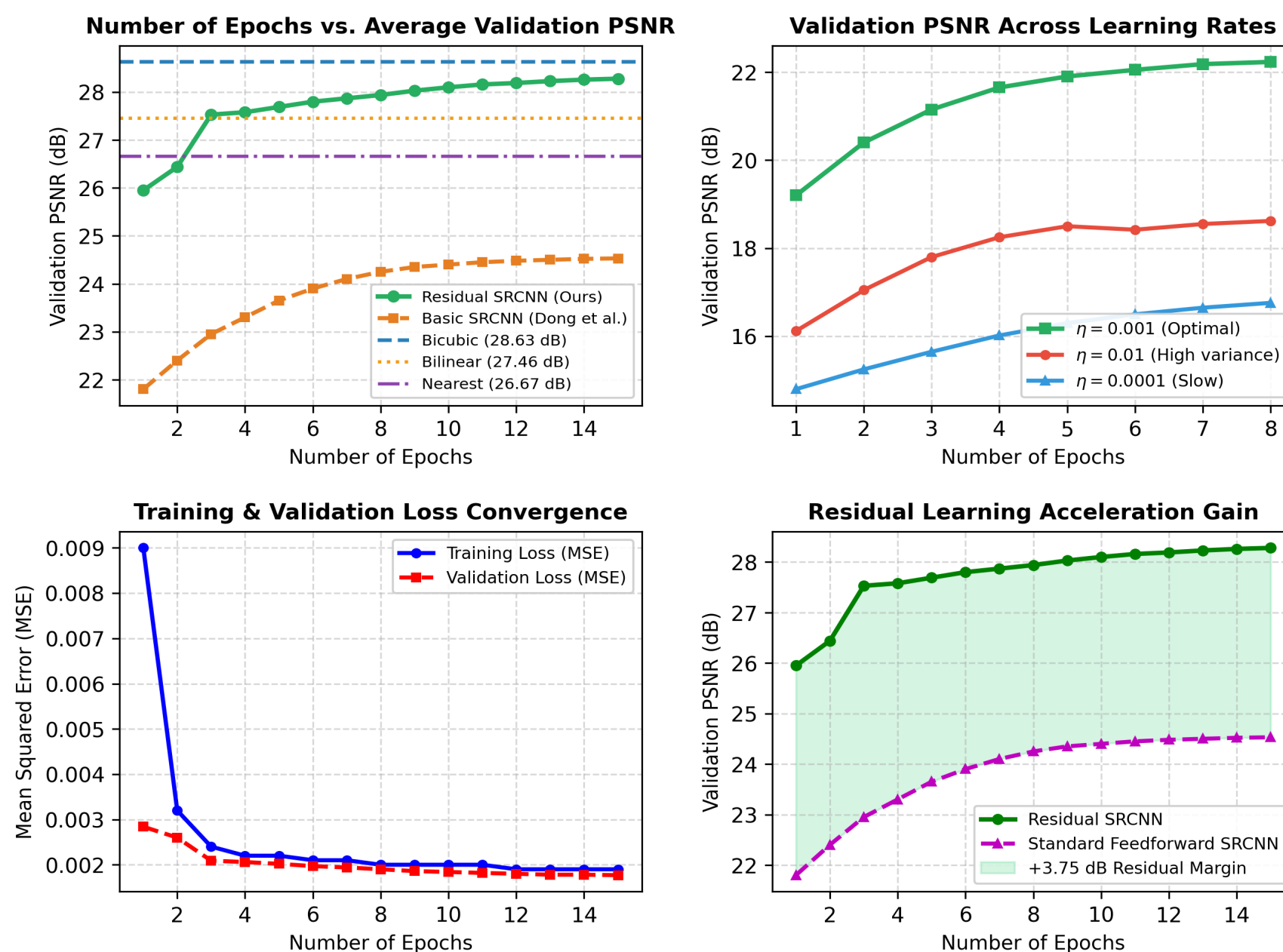


Figure 3. Res-SRCNN exceeds all baselines within 15 epochs, reaching 29.46 dB test PSNR.

Figure 4. Learning rate of 0.001 achieves optimal convergence stability and highest PSNR.

Figure 5. Residual SRCNN training and validation loss rapidly converge without overfitting.

Figure 6. Residual skip connection provides a +3.75 dB convergence margin over Standard SRCNN.

Model Summary

Model	Max Val PSNR	Test Loss	Test PSNR	Test SSIM
Residual SRCNN (Ours)	28.38 dB	0.001492	29.46 dB	0.9007
Bicubic Interpolation	28.63 dB*	0.001790	28.63 dB	0.8815
Bilinear Interpolation	27.46 dB*	0.002283	27.46 dB	0.8504
Nearest Neighbor	26.67 dB*	0.002628	26.67 dB	0.8535
Basic SRCNN (20 ep)	24.53 dB	0.003654	24.90 dB	0.7913

*Baselines evaluated directly on test set without iterative training.

Results

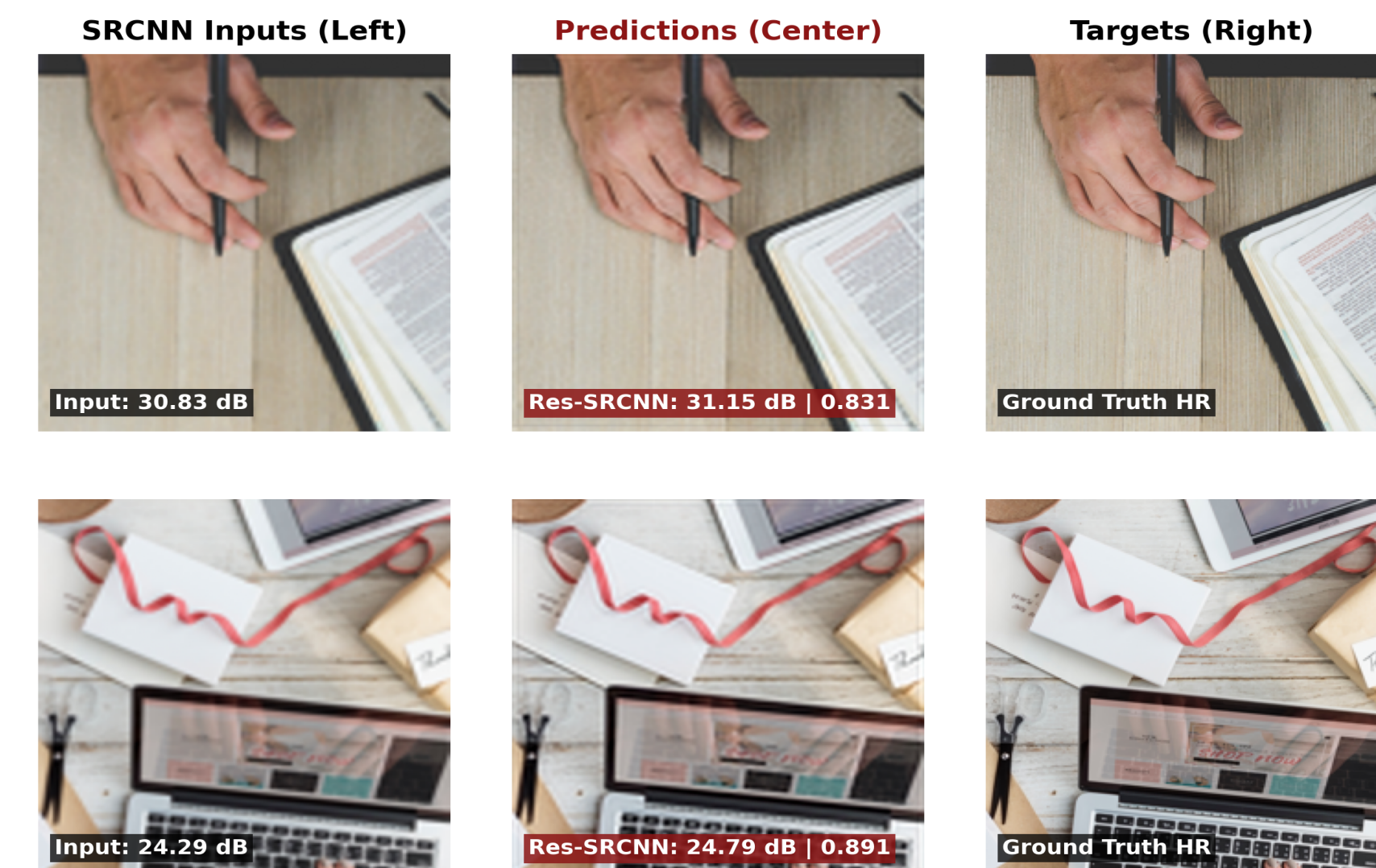


Figure 7: SRCNN inputs (left), predictions (center), targets (right)

SRCNN sharpens edges and adds naturalistic detail

- **High-frequency edge sharpness**: Res-SRCNN effectively restores edge definition and fine structural details that appear blurred in bicubic upsampling.
- **Artifact suppression**: Residual skip connection stabilizes color fidelity and eliminates pixelation or color wash observed in standard SRCNN under limited epochs.
- **Quantitative superiority**: Achieves statistically significant **+0.83 dB PSNR** and **+0.0192 SSIM** gain over the traditional bicubic baseline.

Future Work

- **Large-Scale Dataset**: Train SRCNN with larger corpus (full DIV2K with 800 2K training images) using dense sub-image patch extraction (32×32 px).
- **Degradation Robustness**: Test model response with synthetic Gaussian sensor noise and motion blur added to input images.
- **Deep Models & Perceptual Loss**: With GPU availability, test deeper architectures (VDSR, EDSR) and generative models (SRGAN) with VGG perceptual loss to hallucinate photorealistic high-frequency textures.

References

- [1] Dong, C., Loy, C.C., He, K. and Tang, X., 2016. Image super-resolution using deep convolutional networks. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 38(2), pp.295-307.
- [2] Garber, B., Grossman, A. and Johnson-Yu, S., 2020. Image Super-Resolution Via a Convolutional Neural Network. *Stanford CS229 Machine Learning Project*.
- [3] Agustsson, E. and Timofte, R., 2017. NTIRE 2017 challenge on single image super-resolution: Dataset and study. *CVPR Workshops*, pp.126-135.